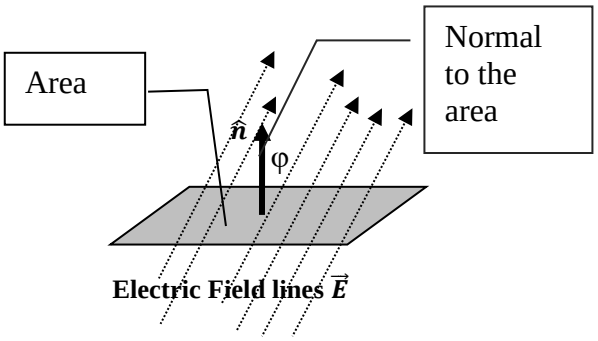
Chapter 36, Electric Fields Part C – Gauss' Law
Calculus Physics II

Gauss' Law, Charge Distributions and the Electric Field - Gauss' Law can be used to formulate the equation of the electric field that is generated by a charge distribution or can determine the net charge of an object by the electric field it generates.

Electric Flux (Φ_E) - is the total Electric field strength that passes through a defined area. The Electric flux is the product of the area times the electric field strength through the inside of that area. The orientation of the area to the electric field can affect that amount of the electric field lines passing through the area so there is an angle dependence. Electric flux has units of Nm^2/C

The angle is measured from a line that is drawn normal (perpendicular) to the surface area relative to the electric field line. The unit vector $\hat{n}$.

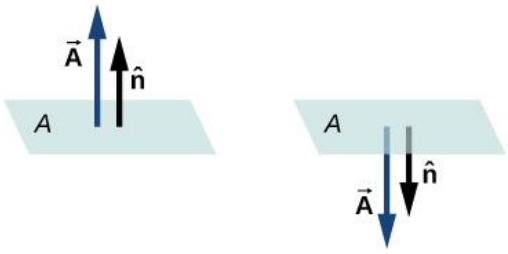


Electric Flux is equal to the dot product of the area vector $\vec{A}$ with the Electric field strength $\vec{E}$ and is a scalar quantity with units of Nm^2/C

$$\Phi_E = \vec{E} \bullet \vec{A}$$

Area vector - For any surface, including a Gaussian surface, the area can be represented by a vector. The magnitude of this vector is equal to the area of the surface element, and its direction is perpendicular (normal) to the surface area.

Since the direction of the area vector is crucial, a unit vector (a vector with a magnitude of 1) is used to specify the direction of the area. In the context of Gauss's Law, this unit vector points outward from the enclosed volume of the Gaussian surface.



$$Area\ vector = \vec{A} = (|\vec{A}|)\hat{n}$$

Where $|\vec{A}|$ is the surface area of the shape and $\hat{n}$ is the unit vector of the area that points outward and is perpendicular to the surface area.

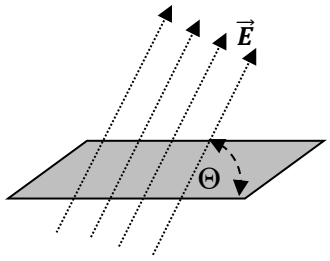
The choice of an outward-pointing normal is a convention, but essential for defining the sign of the electric flux. If the electric field lines are pointing outward through the surface (away from the enclosed charge), the flux is considered positive. Conversely, if the field lines are pointing inward (towards the enclosed charge), the flux is negative.

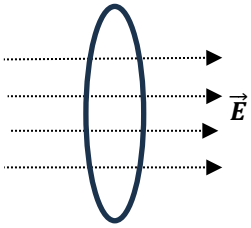
The Electric Flux is then equal to the dot product of the electric field and the area unit vector.

$$\Phi_E = \vec{E} \bullet \vec{A} = \left(|\vec{A}|\right)\vec{E} \bullet \hat{n} = |\vec{E}|Acos(\varphi)$$

Where the Phi φ is the angle between the electric field vector and the area unit vector $\hat{n}$, which is perpendicular to the area.

Example: Determine the electric flux through the surfaces shown below for a uniform electric field of magnitude 200 N/C.

A) 

B) 

C) 

A) Determine the electric flux through a rectangle with a length of 0.2 m and a width of 0.1 m. The electric field makes an angle with the surface area of Θ = 30°.

The angle between the electric field and the normal to the surface is φ = 90° - Θ = 60°

$$\Phi_E = \vec{E} \bullet \vec{A} = |\vec{E}|Acos(\varphi) = (200\ N/C)(0.2m)(0.1m)(\cos(60°)) = 2\ Nm^2/\text{C}$$

B) Determine the electric flux through a hoop with a diameter of 0.5 m with its area perpendicular to the field.

For the hoop A = $\boldsymbol{\pi}$r^2 with r = 0.25 m and φ = 0°

$$\Phi_E = \vec{E} \bullet \vec{A} = |\vec{E}|Acos(\varphi) = (200\ N/C)(0.196\ m^2)(\cos(0°)) = 39.3\ Nm^2/\text{C}$$

C) Determine the electric flux through a hoop with a diameter of 0.5 m with its area parallel to the field.

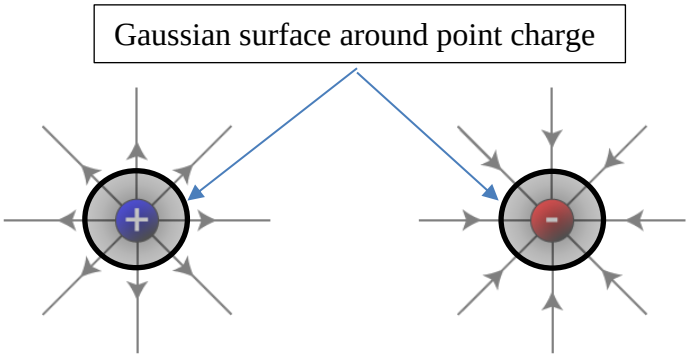
For the hoop A = $\boldsymbol{\pi}$r^2 with r = 0.25 m and φ = 90°

$$\Phi_E = \vec{E} \bullet \vec{A} = |\vec{E}|Acos(\varphi) = (200\ N/C)(0.196\ m^2)(\cos(90°)) = 0\ Nm^2/\text{C}$$

When using Gauss's Law, the area vector for a Gaussian surface is a vector that is perpendicular (normal) to the surface at any given point and points outward from the surface. The unit vector associated with this area vector indicates the direction of this outward normal.

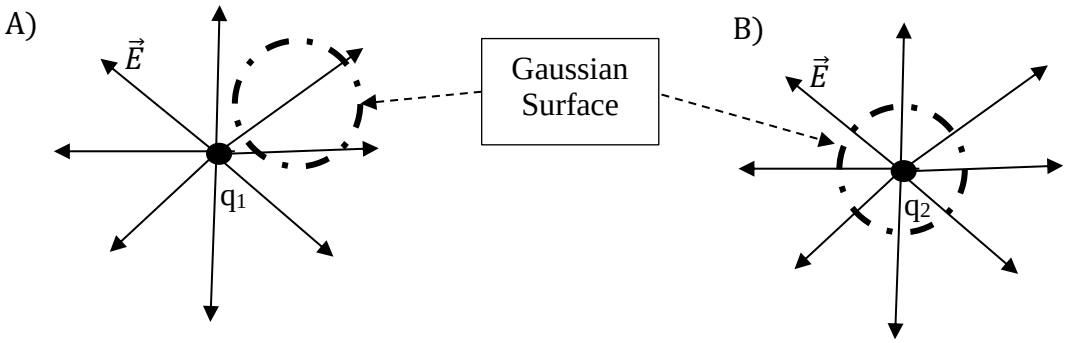
Gaussian Surface: A Gaussian surface is an imaginary closed surface that we can choose to enclose a region of space. This surface is used in conjunction with Gauss's Law to calculate the electric flux through it, which in turn helps determine the electric field due to the enclosed charge distribution.

Gaussian Surface – Theoretical closed surface surrounding the charge distribution.



Note that for the positively charged object the electric field radiates out of the surface and for the negatively charged object the electric field radiates into the object. The type of charge can be determined by how the field enters or exits the Gaussian surface.

Example: Net Flux Through a Gaussian Surface – For two Gaussian Surfaces shown



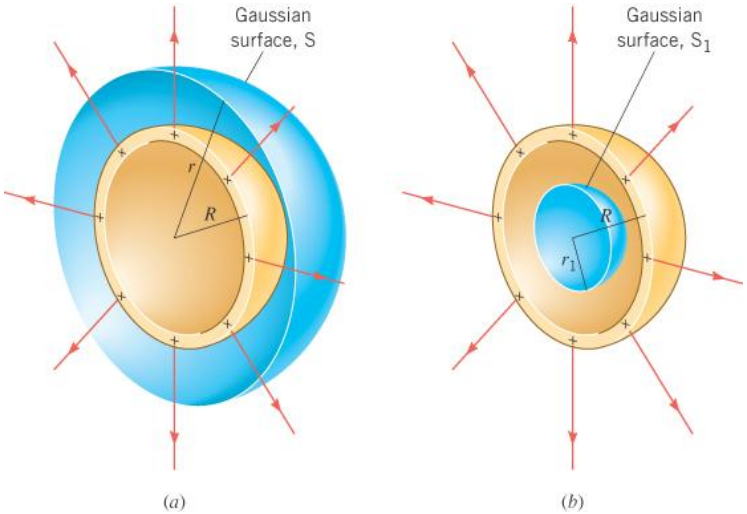
A) The same amount of electric flux is entering the surface as exiting the surface, there is no enclosed charge inside the Gaussian surface therefore there is no net flux through the surface.

For q_1 $\Delta\Phi_E = \Phi_{out} - \Phi_{in} = 0$ Net flux equals zero

B) The electric flux exiting the surface is from the charge inside the surface therefore there is a net electric flux through this Gaussian surface.

For q_2 $\Delta\Phi_E = \Phi_{out} - \Phi_{in} = q_2$ Net flux equals q_2

Example: For a charged conducting sphere shown below (gold sphere).



- For the Gaussian Surface (blue sphere) in (a), the surface includes the charged sphere inside the surface therefore there is a net electric field radiating out of the surface.
- For the Gaussian Surface (blue sphere) in (b) the surface does not include the charged sphere inside the surface therefore there is no net electric field radiating out of the surface.

Electric Flux and a charge distribution – the total Electric field generated by a charge distribution times the surface area it passes through is equal to the electric flux through the surface vector.

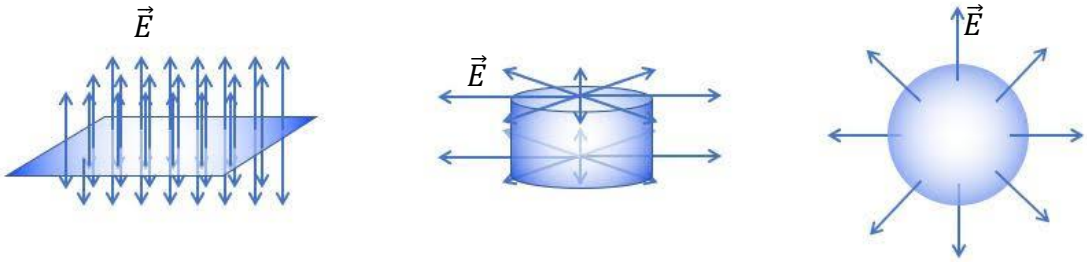
$$\Phi_E = \frac{Q}{\varepsilon_o}$$

$$\Phi_E = \oint \vec{E} \bullet \hat{n} dA$$

The surface integral $\oint$ in Gauss's law: The surface integral of the electric field is evaluated over a closed surface. This closed surface, often referred to as a Gaussian surface, can be any arbitrary shape, but is strategically chosen to simplify calculations based on symmetry. The integral essentially calculates the total electric flux passing through this closed surface, which is directly related to the net electric charge enclosed within it.

If the Gaussian Surface is chosen so that the electric field is everywhere perpendicular to the surface and the equation for this surface area is known, then the integration is not required. The Gaussian Surface equation (square, cylinder or sphere) is the integration.

Gaussian surfaces and the electric field



Gauss's Law states that the surface integral of the electric field ($\vec{E}$) over any closed surface is proportional to the total electric charge enclosed (Q) within that surface, where dA is an infinitesimal area element on the surface, and ε_0 is the permittivity of free space.

$$\Phi_E = \oint \vec{E} \bullet \hat{n} dA = \frac{Q_{in}}{\varepsilon_o}$$

Where

Φ_E – The electric flux through a surface
Q_{in} – total net charge inside surface.
$\vec{E}$ – Electric field strength vector
ε_o - Permittivity of free space and is equal to 8.85 x 10^{-12} C^2/Nm^2
$\hat{n}$ – area unit vector normal to the surface area.
dA – incremental surface area.
$\vec{E} \bullet \hat{n} = |\vec{E}| \cos(\varphi)$ is the dot product
where φ is the angle between the electric field and the normal to the surface area $\hat{n}$.

- The key requirement is that the surface must be closed, meaning it completely encloses a volume. Common examples include spheres, cubes, and cylinders.
- The choice of the Gaussian surface is crucial for simplifying the calculation. By exploiting symmetry (spherical, cylindrical, or plane), the electric field can be made constant and perpendicular to the surface, the integral simplifies to the electric field times the area of the Gaussian surface.

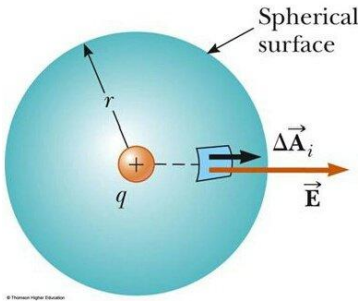
<u>Solving Problems using Gauss's Law</u>

1) Pick a Gaussian Surface that makes the geometry and equation simple so that the electric field is perpendicular to the surface at all points and is at the distance you want to know the magnitude of the Electric Field. The electric field lines MUST be perpendicular to the surface at all point to apply Gauss's Law to a problem.
2) Make sure that the surface includes all of the charge distribution you are interested in.
3) Determine the area of the Gaussian Surface and the total charge enclosed by the surface.
4) Make sure you Sum all the Gaussian surfaces that make up the total surface.
5) Solve for the Electric Field.

Example: For a point charge, a spherical Gaussian surface centered on the charge simplifies the calculation. Since the electric field is radial and has the same magnitude at all points on the sphere, the surface integral becomes E * $4\pi r^2$, where r is the radius of the sphere. The integral is solved using the known area of the sphere.

$$\Phi_E = \oint \vec{E} \bullet \hat{n} dA = \oint |\vec{E}| \cos(0°) dA = |\vec{E}| \oint dA = |\vec{E}| A_{sphere} = |\vec{E}| (4\pi r^2)$$

$$\Phi_E = \frac{Q_{in}}{\varepsilon_o} = |\vec{E}| (4\pi r^2)$$

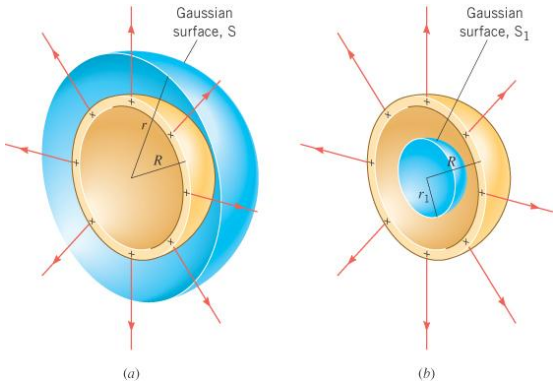


Solving for the electric field gives

$$|\vec{E}| = \frac{1}{4\pi\varepsilon_o}\left(\frac{Q_{in}}{r^2}\right)$$

- This is the equation for the electric field strength r distance from a point charge.

Example: A hollow conducting sphere with radius R is shown below. Determine the electric field strength for the region enclosed by the Gaussian surfaces first shown outside the sphere and them inside the sphere.



(a) The area of the Gaussian sphere (blue sphere) is given is equal to $A_{sphere} = 4\pi r^2$ and the charge on the sphere is given by the surface charge and is $Q_{in} = \sigma A = \sigma(4\pi R^2)$.

$$\Phi_E = \oint \vec{E} \bullet \hat{n} dA = \frac{Q_{in}}{\varepsilon_o} = \oint |\vec{E}| \cos(0°) dA = |\vec{E}| A_{sphere} = |\vec{E}| (4\pi r^2)$$

$\frac{Q_{in}}{\varepsilon_o} = |\vec{E}| (4\pi r^2)$ Solving for the electric field and subsituding for Q_{in}

$$|\vec{E}| = \frac{Q_{in}}{(4\pi r^2)\varepsilon_o} = \frac{\sigma 4\pi R^2}{(4\pi r^2)\varepsilon_o} = \frac{\sigma R^2}{r^2 \varepsilon_o}$$

(b) For a conductor the charge resides on the surface of the conductor, the Gaussian surface shown inside the conducting sphere has no charge inside the Gaussian surface (blue sphere) therefore Q_{in} = 0 C.

$$\Phi_E = \left|\vec{E}\right| A_{sphere} = \frac{Q}{\varepsilon_o} = \frac{0\ C}{\varepsilon_o} = 0$$

$$\left|\vec{E}\right| = 0\ N/C$$

- There is no electric field inside of a conductor!

Example: A solid conducting sphere has a net charge, if the electric field strength r = 0.4 m from the center of the sphere has a magnitude of 500 N/C, determine the net charge on the sphere.

$$\Phi_E = \left|\vec{E}\right| A_{sphere} = \frac{Q}{\varepsilon_o}$$

Place the spherical Gaussian surface at r = 0.4 m and solve for Q

$$Q = \varepsilon_o \left|\vec{E}\right| (4\pi r^2) = 8.9\ x\ 10^{-9} C = 8.9\ nC$$

Example: Given solid non-conducting sphere of radius R and with a volume charge density of ρ with the charge distributed uniformly thought the volume.

A) Determine an equation for the total net charge of the sphere.

$$\rho = \frac{Q_{Total}}{V} \quad Solve\ for\ Q$$

$$Q_{Total} = \rho V = \rho \left[\frac{4}{3}\pi R^3\right]$$

B) Determine an equation for the electric field strength at a point a distance r_{out} that is larger than the radius of the sphere (r_{out} > R)

$$\left|\vec{E}\right| = \frac{Q_{in}}{A\varepsilon_o} = \frac{\rho\left[\frac{4}{3}\pi R^3\right]}{(4\pi r_{out}^2)\varepsilon_o} = \frac{\rho R^3}{3r_{out}^2\varepsilon_o}$$

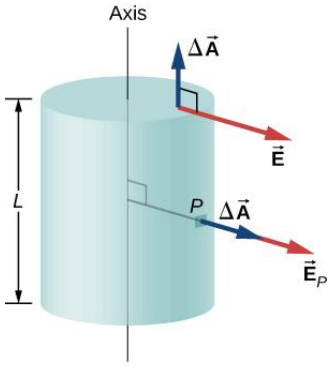
C) Determine an equation for the electric field strength at a point a distance r_{in} that is smaller than the radius of the sphere ($r_{in} < R$)

$$Q_{in} = \rho V_{Gausian} = \rho\left[\frac{4\pi r_{in}^3}{3}\right]$$

$$|\vec{E}| = \frac{Q_{in}}{A\varepsilon_o} = \frac{\rho\left[\frac{4}{3}\pi r_{in}^3\right]}{(4\pi r_{in}^2)\varepsilon_o} = \frac{\rho r_{in}}{3\varepsilon_o}$$

Example: A solid non-conducting cylinder has a uniform volume charge density of ρ a length of L and a radius of R.

Note that the electric field is perpendicular to the cylinder's surface but is parallel to the cylinder's ends. This means that the electric flux is zero on the ends and we will only need to consider the surface as our Gaussian surface.



Volume of a cylinders is $V_{cylinder} = \pi R^2 L$

Area of the surface of the cylinder is $A = 2\pi RL$

A) Determine an equation for the total net charge of the cylinder.

$$\rho = \frac{Q_{Total}}{V} \quad Solve\ for\ Q$$

$$Q_{Total} = \rho V = \rho[\pi R^2 L]$$

B) Determine an equation for the electric field strength at a point a distance r_{out} that is larger than the radius of the cylinder ($r_{out} > R$)

$$|\vec{E}| = \frac{Q_{in}}{A\varepsilon_o} = \frac{\rho[\pi R^2 L]}{(2\pi r_{out} L)\varepsilon_o} = \frac{\rho R^2}{2r_{out}\varepsilon_o}$$

C) Determine an equation for the electric field strength at a point a distance r_{in} that is smaller than the radius of the cylinder ($r_{in} < R$)

$$Q_{in} = \rho V_{Gausian} = \rho[\pi r_{in}^2 L]$$

$$|\vec{E}| = \frac{Q_{in}}{A\varepsilon_o} = \frac{\rho[\pi r_{in}^2 L]}{(2\pi r_{in} L)\varepsilon_o} = \frac{\rho r_{in}}{2\varepsilon_o}$$